



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester II - UCSC AY20 [held in March/ April 2024]

SCS 1213 – Probability and Statistics

(Two (2) Hours)

Answer ALL questions

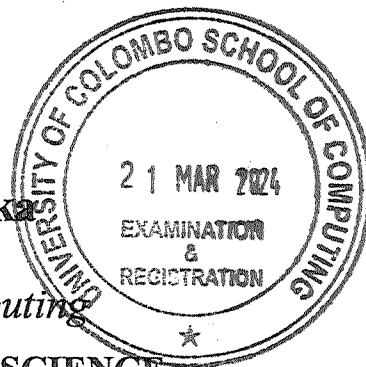
Number of Pages = 09

Number of Questions = 04

To be completed by the candidate

Index Number:

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32

Important Instructions to candidates:

- I. Question number 01 and 02 encircle the correct choice in the tables given in the page 02.
- II. Students should answer to the question number 03 and 04 in the medium of English language only using answer booklets.
- III. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- IV. Write your index number CLEARLY on each and every page of this Question paper and answer booklets.
- V. Attach this question paper with answer booklets.
- VI. This paper consists of 04 questions in 09 pages (including the Cover Page).
- VII. Answer ALL questions.
- VIII. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- IX. Non-Programmable calculators are allowed.
- X. Do not tear off any part of this question paper. Under no circumstances may this question paper, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

Answers Table for Question Number 01

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice in the following table.

Question Number	Answers					Question Number	Answers				
i	a	b	c	d	e	ix	a	b	c	d	e
ii	a	b	c	d	e	x	a	b	c	d	e
iii	a	b	c	d	e	xi	a	b	c	d	e
iv	a	b	c	d	e	xii	a	b	c	d	e
v	a	b	c	d	e	xiii	a	b	c	d	e
vi	a	b	c	d	e	xiv	a	b	c	d	e
vii	a	b	c	d	e	xv	a	b	c	d	e
viii	a	b	c	d	e						

Answers Table for Question Number 02

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice in the following table.

Question Number	Answers					Question Number	Answers				
i	a	b	c	d	e	ix	a	b	c	d	e
ii	a	b	c	d	e	x	a	b	c	d	e
iii	a	b	c	d	e	xi	a	b	c	d	e
iv	a	b	c	d	e	xii	a	b	c	d	e
v	a	b	c	d	e	xiii	a	b	c	d	e
vi	a	b	c	d	e	xiv	a	b	c	d	e
vii	a	b	c	d	e	xv	a	b	c	d	e
viii	a	b	c	d	e						

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice on the 2nd page of this question paper.

- 3

Question Number 02 (30 Marks)

This question consists of 15 multiple choice questions. Select the most suitable answer and encircle the correct choice on the 2nd page of this question paper.

- (i). Let A, B, C and D be four events. Which of the following represents the event that none of the four events occurs?
- a). $(A \cap B \cap C \cap D)^c$ c). $A^c \cap B^c \cap C^c \cap D^c$ e). $A^c \cap B^c \cup C^c \cap D^c$
b). $(A \cup B \cup C \cup D)^c$ d). $A^c \cup B^c \cup C^c \cup D^c$
- (ii). If $P(A) = 0.6$, $P(B^c) = 0.4$ and $P(B | A) = 0.4$ then $P(A | B)$ is:
- a). 0.26 b). 0.40 c). 0.50 d). 0.60 e). 0.80
- (iii). If $P(A) = 0.4$, $P(B | A) = 0.3$ and $P(A | B) = 0.4$ then $P(B^c)$ is:
- a). 0.06 b). 0.26 c). 0.30 d). 0.60 e). 0.70
- (iv). If $P(A|B) = 0.6$, $P(B) = 0.4$ and $P(A \cup B) = 0.6$, then $P(A)$ is:
- a). 0.14 b). 0.24 c). 0.34 d). 0.44 e). 0.54
- (v). If events A and B are independent, then always:
- a). $P(A \cup B) = P(A) + P(B)$ d). $P(A \cup B) = P(A) \cup P(B)$
b). $P(A \cup B) = P(A) \times P(B)$ e). $P(A \cap B) = P(A) + P(B)$
c). $P(A \cap B) = P(A) \times P(B)$
- (vi). The mean of a binomial distribution with parameters n and p is:
- a). $n(1-p)$ b). $np(1-p)$ c). $(n-1)p$ d). np e). $p(1-p)$
- (vii). The standard deviation of a binomial distribution with the values of parameters $n= 16$ and $p = 0.1$ is:
- a). 1.20 b). 1.26 c). 1.44 d). 1.60 e). 1.88
- (viii). A box contains four red balls and five blue balls. When two balls are selected without replacement, the probability that at least one red ball is;
- a). 20/72 b). 25/72 c). 52/72 d). 25/81 e). 50/81
- (ix). The probability of an impossible event is,
- a). 1 b). -1 c). 0.1 d). 0 e). 0.5

(x). Suppose that 0.1% students are infected with H1N1 virus flu in a certain university. Let Y be the number of students with H1N1 virus flu in a sample of size 250. The exact distribution of Y is;

- a). Binomial with $n = 250$ and $p = 0.1$
- b). Binomial with $n = 250$ and $p = 0.01$
- c). Binomial with $n = 250$ and $p = 0.001$
- d). Poisson with $\lambda = 0.1$
- e). Poisson with $\lambda = 25$

(xi). If $P(A|B) = 1$ then;

- a). A and B are impossible events.
- b). A and B are mutually exclusive.
- c). $P(A \cap B) = P(A)P(B)$
- d). $A \subset B$
- e). $B \subset A$

(xii). A committee of two executives is randomly selected from a group of three male and five female executives. The probability that the committee contains at least one female:

- a). $\frac{{}^3C_0 {}^5C_2}{{}^8C_2}$
- b). $1 - \frac{{}^3C_2 {}^5C_0}{{}^8C_2}$
- c). $\frac{{}^3C_2 {}^5C_0}{{}^8C_2} - 1$
- d). $1 - \frac{{}^3C_0 {}^5C_2}{{}^8C_2}$
- e). $\frac{{}^3C_0 {}^5C_2}{{}^8C_2} - 1$

(xiii). If the standard deviation of a Poisson distribution is 3 then the mean is:

- a). 1.50
- b). 1.73
- c). 3.00
- d). 6.00
- e). 9.00

(xiv). Let X be a random variable with $E[X] = 6$ and $V[X] = 4$. Then the expected value and the standard deviation of $Y = 3X - 1$ are respectively:

- a). 11, 12
- b). 17, 36
- c). 17, 6
- d). 18, 6
- e). 18, 36

(xv). Which one of the following does not represent a valid probability distribution function?

- a). $P[X = x] = (x + 2)/25, \quad x = 1, 2, 3, 4, 5$
- b). $P[X = x] = (x + 2)/15, \quad x = -1, 0, 1, 2, 3$
- c). $P[X = x] = (x - 2)/5, \quad x = 1, 2, 3, 4, 5$
- d). $P[X = x] = 1/5, \quad x = -2, -1, 0, 1, 2$
- e). $P[X = x] = x/15, \quad x = 1, 2, 3, 4, 5$

Question Number 03 (20 Marks)

(a). A manufacturer of LED televisions encountered a shortage of LED screens, since the usual supplier sends only 65% of LED screens from the total requirement. Shipments from that usual supplier had 4% defective screens. Because of the shortage, the manufacturer had to obtain 20% of the LED screens from a second supplier who sends 7% defective screens, and the remaining 15% from a third supplier who sends 10% defective screens to fulfil his total requirement. If a LED screen was found to be defective what is the probability that it was sent by the second supplier.

(b). The probability density function of a discrete random variable X is given by the function

$$f(x) = k \binom{5}{x} \binom{4}{3-x} \quad \text{for } x = 0, 1, 2, 3$$

- (i). Calculate the value of k .
- (ii). Calculate the mean of X

Question Number 04 (20 Marks)

(a). Suppose 20% of the Teak trees in a forest have severe leaf damage due to air pollution. Twenty Teak trees are selected randomly from that forest. Calculate the probability that at least 1 Teak tree have leaf damage problem.

(b). The average number of breakdowns per week in a computing system of a certain office is 0.4.

- (i). What is the most suitable probability distribution to model the number of breakdowns per week? Give reasons.
- (ii). Calculate the probability that the computing system will run for two consecutive weeks without breakdown

(c). A statistical analysis of International Direct Dialing (IDD) calls made from a global marketing company in Sri Lanka indicates that the length of these calls is normally distributed, with average 240 seconds and standard deviation 40 seconds. If the company is planning to contact 100 international customers in the next month through IDD calls,

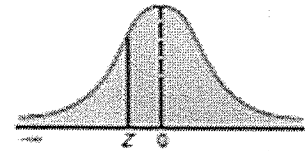
- (i). Determine the expected number of IDD calls with the duration of a call between 110 and 180 seconds
- (ii). What is the cut-off time point for the longest 5 % IDD calls?

***** The End of the Question Paper *****

TABLE E.2

The Cumulative Standardized Normal Distribution

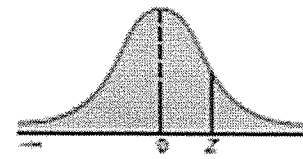
Entry represents area under the cumulative standardized normal distribution from $-\infty$ to Z



Cumulative Probabilities										
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-6.0	0.000000001									
-5.5	0.000000019									
-5.0	0.000000287									
-4.5	0.000001398									
-4.0	0.000031671									
-3.9	0.00003	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0733	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2031	0.2003	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

TABLE E.2

The Cumulative Standard Normal Distribution (continued)

Entry represents area under the cumulative standardized normal distribution from $-\infty$ to Z .

Cumulative Probabilities										
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5198	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99897	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99987	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4.0	0.999968329									
4.5	0.999996602									
5.0	0.999999713									
5.5	0.999999981									
6.0	0.999999999									

